ReformAI: Model Strategy & System Architecture

I'm grounding this in current model capabilities and vendor docs, then mapping them to ReformAI's control requirements to propose an architecture and phased rollout with explicit tradeoffs, not a surface-level winner-takes-all model choice.

The early read is now consistent: prompt-native APIs from Google and OpenAI have improved on multi-turn edits, masking, and reference use, but they still don't provide geometry-guaranteed control. I'm now validating the controllable/open-stack options and which APIs actually expose masks, structure references, and multi-reference inputs, because that boundary likely determines the architecture.

I've got enough to make the architecture call: no single current model can satisfy ReformAI's hard constraints, so the recommendation is converging on a two-lane system. Use a fast conversational model for ideation and iterative UX, and a controlled diffusion/compositing lane—likely Stability-style ControlNet plus masks, segmentation, reranking, and object-aware product workflows—for geometry-locked, SKU-faithful edits.

Thought for 31m 51s

ReformAI should not make this as a "pick the best image model" decision. The right answer is a multi-model, constraint-aware editing system in which the generator is only one component. Today's strongest closed models now support multi-turn image editing, multiple references, masking, and higher-fidelity instruction following, while open/composable stacks expose the structural controls that matter for room preservation. But no single public model currently gives you all of these at once: reliable multi-turn editing, explicit masked/local edits, depth/edge conditioning, immutable-region preservation, product identity control, and commerce-grade validation.

A. Model Landscape Overview

OpenAI: GPT Image / gpt-image-2 (with DALL·E now secondary).
OpenAI's current image stack is GPT Image, not legacy DALL·E. The Image API supports edits from one or more source images, up to 16 input images for GPT image models, mask-based edits, and an input_fidelity control; the Responses API adds multi-turn conversational editing. OpenAI also positions gpt-image-2 as its most capable image model so far, with stronger editing, layout, text rendering, and instruction following. This makes OpenAI very good for conversational editing UX and multi-image prompt/reference workflows. The weakness is that the public image APIs are still fundamentally prompt-native: they do not expose first-class depth/edge/segmentation conditioning in the way ControlNet-style systems do. For ReformAI, that means OpenAI is strong for ideation, controlled restyling attempts, and conversational refinement, but not sufficient by itself for strict camera/layout preservation. Supports: img2img/editing yes; masking yes; inpainting yes; segmentation no; depth/edge conditioning no; multi-image conditioning yes.

Google: Gemini image models are strong conversational editors, but the full Google answer is Gemini + Imagen controls.
Gemini 2.5 Flash Image is explicitly optimized for price/performance and fast image generation/editing, supports multi-turn image editing, and works best with up to three input images. Gemini 3.1 Flash Image and Gemini 3 Pro Image push further: Google positions 3.1 Flash for speed/high-volume and 3 Pro for professional asset production and complex multi-turn workflows; Gemini 3 image models support up to 14 references, with separate limits for objects and characters. That makes Google currently one of the best ecosystems for multi-image blending and conversational edits. The catch is

that Gemini by itself is still mostly a prompt-driven editing surface. Google's broader Imagen editing/customization stack does add masks, automatic segmentation, inpainting, background replacement, and canny/scribble controlled customization, but Google's own docs also note limitations for some combinations of style reference plus control images. So Google is strong overall, but only if ReformAI uses more than the current Gemini-only prompt flow. Supports: img2img/editing yes; masking/inpainting yes in the broader Imagen stack; automatic segmentation/background workflows yes; edge conditioning yes via controlled customization; multi-image conditioning yes.

Stability AI / Stable Diffusion ecosystem: the best current path for explicit spatial control.
Stability's current platform is organized around Generate, Edit, and Control services. Its docs explicitly call out mask-based inpainting, product-placement-oriented edit tools, background removal/replacement, relighting, and Control services built on ControlNets and similar technologies. Stability also released SD 3.5 Large ControlNets for Blur, Canny, and Depth; its own documentation says Canny structures images using edge maps, and Depth is useful for architectural renderings and exact composition control. This is exactly the kind of control ReformAI needs for geometry-locked room editing. The downside is engineering burden: you gain controllability, but you own more of the pipeline, tuning, and failure handling. Supports: img2img yes; masking/inpainting yes; structural conditioning yes; depth/edge conditioning yes; segmentation via external tooling, not native scene understanding.

Black Forest Labs / FLUX: the strongest current "editor-first" model family for a product like ReformAI.
BFL explicitly recommends FLUX.2 for new projects and recommends FLUX.2 [PRO] for editing workflows. Its docs and model pages position FLUX.2 as strong on multi-reference editing, photorealism, product visualization, reliable spatial reasoning, pose guidance, and consistency; BFL also markets FLUX.2 for retexturing while preserving shape, geometry, and lighting, and FLUX.1 Fill [pro] exists as a dedicated image+mask fill endpoint. This makes FLUX.2 the most compelling current primary editing engine for ReformAI's medium-term stack. The limitation is the same one that applies to every general-purpose editor: FLUX.2 can improve consistency, but it still does not magically make constraints enforceable without masks, control maps, and validation. Supports: editing yes; multi-reference yes (up to 10 images); masked fill yes via Fill endpoint; explicit depth/edge control not first-class in the main FLUX.2 API surface, so pair it with external structure controls or specialized tools.

Adobe Firefly: best positioned for enterprise-safe product workflows and brand/product consistency.
Adobe's Firefly APIs expose Structure Reference, which uses outline and depth from a reference image, plus Style Reference, plus Custom Models that can be trained on products, characters, or styles for brand consistency. Adobe also exposes composite/object-composite workflows for blending product shots into scenes with harmonized tones, colors, lighting, shadows, and textures. For ReformAI, that makes Firefly especially relevant to the commerce-grade SKU visualization lane, where exact product appearance, enterprise/legal posture, and compositing matter more than raw generative freedom. The downside is lower openness and less fine-grained low-level control than the best open diffusion stacks. Supports: structure reference yes; style reference yes; custom product/brand models yes; composite/product workflows yes; lower composability than SD/ControlNet ecosystems.

Midjourney: strong for inspiration, weak for deterministic renovation editing.
Midjourney's own docs say Style References capture vibe rather than copying objects or people, and Image Prompts are inspiration for a new image rather than precise copying. Its editor does support masks, erase-based edits, layers, and a Retexture tool that keeps structure/composition while changing style. That makes Midjourney useful for moodboard exploration or "what vibe do we want?" workflows, but not a sensible backbone for ReformAI's constraint-aware editing system. Supports: references yes; masking yes in editor; local edits yes; explicit structural conditioning no; deterministic product/room preservation no.

Runway Gen-4 Image: useful creative reference system, not a room-preservation backbone.
Runway positions Gen-4 Image as a high-fidelity base model with references and iterative workflows. Its References system supports up to three references and even sketch references to influence placement/composition. But Runway's "Vary" feature intentionally explores different structures and compositions, which is the opposite of ReformAI's hardest constraint. It is better suited to concept exploration than strict controlled renovation transforms. Supports: image references yes; sketch references yes; multi-reference yes (up to 3); explicit masking/structural locks are not the center of the product.

Segmentation and control tools are not optional for ReformAI.
SAM 2 is a promptable segmentation foundation model; Grounding DINO is an open-set detector that can find arbitrary categories from text and is already combined with SAM 2 in Grounded SAM 2; Florence-2 is a unified prompt-based vision model for captioning, detection, grounding, and segmentation. ControlNet adds spatial conditioning to diffusion models using controls such as edges, depth, segmentation, and pose. MiDaS provides monocular depth from a single image, and IP-Adapter gives diffusion models a lightweight image-prompt pathway. These are the pieces that convert "creative image editing" into "constraint-aware image editing."

Emerging specialist research is relevant, but as adjuncts rather than the primary stack.
AnyDoor is built for zero-shot object-level insertion at user-specified locations; PowerPaint is designed for versatile inpainting including shape-guided insertion and outpainting; BrushNet is a plug-and-play inpainting model; GLIGEN grounds generation using explicit grounding signals such as boxes. These are valuable R&D; modules for product insertion and high-quality local edits, but they are better treated as specialist plugins after ReformAI establishes its core controlled pipeline.

B. Capability Mapping to ReformAI's Requirements

1. Spatial preservation (geometry, camera, perspective).
This is the requirement that most clearly rules out a pure prompt-only architecture. The most realistic current solutions are structure-conditioned diffusion pipelines: SD 3.5 + Canny/Depth ControlNets, Google's controlled customization with canny/scribble inputs, and Adobe Structure Reference. FLUX.2 can be part of this lane, but not as the only mechanism. OpenAI GPT Image, Gemini-only prompting, Midjourney, and Runway can approximate the original room, but they do not expose the hard structural levers needed for reliable camera/layout preservation.

2. Window/background preservation.
This is not a "better prompt" problem. It requires explicit protected masks around windows and outside views, local editing only outside those masks, and post-edit validation on the preserved crops. Google's Imagen docs and Stability's edit stack both support mask-driven edits; OpenAI and FLUX also support masked/image edits. But the exact preservation of the exterior view is something ReformAI should enforce by system design, not leave to the generator. Prompt-only scene editors will drift on window shape, brightness, and exterior content.

3. Controlled transformation vs. regeneration.
Models that expose masks, inpainting, or image-to-image edit surfaces can participate in controlled transformation: OpenAI image edits, FLUX.1 Fill / FLUX.2 editing, Stability inpainting/edit endpoints, Google Imagen inpainting/background replacement, and Adobe Instruct Edit/composites. Prompt-only room restyling in Gemini, Midjourney, Runway, or even GPT Image without masks tends to behave like partial regeneration of the scene rather than a constrained transform. ReformAI's current Gemini 2.5 Flash Image-only prompt flow sits on the wrong side of that boundary.

4. Furniture and product identity preservation.
Current models can improve identity preservation, but not guarantee it. FLUX.2 is strong here because of multi-reference editing and BFL's product-visualization emphasis; Gemini 3 models support many reference images; OpenAI supports many input images and high-fidelity multi-turn edits; Adobe Custom Models can train on products/characters for consistency. But exact SKU fidelity still requires explicit product assets and validation. A generator alone should not be the source of truth for manufacturer-specific geometry, stitching, logos, or finish variants.

5. Multi-input blending (base image + style + moodboard + products).
Google is especially strong here because Gemini 3 supports up to 14 references; OpenAI allows up to 16 input images; FLUX.2 supports up to 10 references; Runway supports up to 3; Midjourney supports multiple references for inspiration; Firefly has both structure and style references. But blending many inputs under hard room constraints is exactly where prompt-native systems become unstable. ReformAI should treat multi-input blending as a planned, weighted edit specification, not as "throw all the images into one prompt."

6. Region-based editing (future requirement).
This is feasible now, but only with masks. OpenAI image edits support masks; Google Imagen supports masked inpainting and masked object removal/insertion; Stability supports inpainting/edit flows; FLUX has a dedicated Fill endpoint; Midjourney's editor has masks, though it is not suited as a platform backbone. Region-based editing should be treated as a product/UI capability backed by a mask service, not as a feature that magically appears once you change the image model.

7. Iterative refinement and stateful sessions.
OpenAI's Responses API and Google's Gemini 3 image flows both explicitly support multi-turn editing, which is useful for natural language refinement. But "stateful sessions" in ReformAI's sense still belong to the application: you must persist the accepted image, masks, control maps, edit history, products, and validation scores. The model helps with conversational continuity; it does not replace state management.

8. Commerce-grade product rendering.
This is where pure generation is weakest. Adobe is strongest on explicit product/brand workflows through Custom Models and Object Composites. FLUX.2 is promising for product visualization and identity-preserving edits. Google and OpenAI can help with contextual integration. Research systems like AnyDoor and PowerPaint are relevant for insertion and local harmonization. But if the requirement is truly commerce-grade SKU fidelity, ReformAI should use explicit product assets—2D cutouts at minimum, ideally 3D/CAD/PBR when available—and use generative models for harmonization, relighting, and scene integration rather than for inventing the SKU itself.

C. Prompt-Based vs Controlled Pipeline Analysis

Where prompt-based systems break down.
Prompt-first models are good at "make this room Scandinavian" or "add warmer lighting," but they fail when the user implicitly means "do that while preserving my exact camera position, walls, windows, exterior view, floor boundaries, and untouched furniture." Without masks or structural conditions, the model is free to reinterpret the scene. That produces the failure mode ReformAI already sees: outputs that feel like scene generation rather than constrained transformation.

What controlled pipelines add.
Controlled pipelines add three things prompt-only systems do not give reliably: first, protected regions via masks; second, structure anchoring via depth, canny, or structure references; third, automated

acceptance criteria via validation/reranking. This is what turns “a nice model” into “a product that can preserve windows, geometry, and products on demand.”

When prompt-driven systems still matter.
They still matter for ideation, moodboard expansion, inspiration, conversational UX, and rapid restyle exploration. Gemini 3, GPT Image, Midjourney, and Runway are all useful in that lane. ReformAI should keep a prompt-native path—but explicitly as Creative Mode, not as the default engine for locked room edits.

The right product split.
ReformAI should have at least two operating modes: Creative Mode for broad stylistic exploration, and Locked Edit Mode for real renovations where structure and product constraints matter. Trying to force one prompt-driven path to serve both is the architectural mistake to avoid.

D. Recommended Model Strategy (Single vs Multi)

Single model? No.
A single-model architecture is not the right choice for ReformAI’s product vision. The required capabilities are spread across different systems: high-quality editing, segmentation/masking, structural conditioning, validation, and product compositing. Even the best current models only cover subsets of that stack.

Recommended: multi-model architecture.
The best strategy is:

Model A: Primary editor/generator — FLUX.2 [PRO] by default; FLUX.2 [MAX] for premium quality.
Best current choice for identity-aware editing, multi-reference workflows, and production editing.
Model B: Geometry-lock path — SD 3.5 Large + Canny/Depth ControlNets.
Use when spatial preservation is non-negotiable. This is the structural backbone prompt-only models lack.
Model C: Segmentation / mask generation — SAM 2 + Grounding DINO or Florence-2.
Needed to identify windows, walls, floors, preserved furniture, and editable zones.
Model D: Local inpainting / fill — FLUX.1 Fill [pro] or Stability inpainting.
Use for region edits, product swaps, cleanup, and controlled local insertions.
Model E: Validator / reranker — a strong VLM judge + CLIP/DINOv2/LPIPS metrics.
Use semantic alignment scoring plus hard preservation scoring on protected regions and product identity.
Model F: Commerce product lane — deterministic product asset compositor, with Adobe Firefly Object Composites/Custom Models as an enterprise option.
Use explicit product assets as ground truth; generative models then harmonize them into the room.

If you force a single primary image model anyway: choose FLUX.2 [PRO] as the default editor, not Gemini 2.5 Flash Image. FLUX.2 is the best single primary editing family here because BFL is explicitly steering new projects to FLUX.2, positions it as the recommended editing model, and supports multi-reference editing with stronger spatial/product-focused claims than the current prompt-only Gemini stack. But even then, you still need masks, control maps, and validators around it.

If vendor consolidation matters more than best-in-class control: the strongest single-vendor compromise is Google, because it combines Gemini conversational image editing with Imagen masked editing, background replacement, and controlled customization. But it is still a compromise, and Google’s own docs call out limitations in some controlled-customization use cases.

E. Proposed Target Architecture

1) Preprocessing / scene parsing.
Use SAM 2 for promptable segmentation, Grounding DINO or Florence-2 to label windows, walls, floor, ceiling, doors, sofas, tables, and other editable/non-editable objects, and MiDaS plus line/edge extraction to create depth and edge maps. Persist these artifacts as part of session state; they are reusable across edits in the same room.

2) Edit planner / constraint builder.
Translate the user request into an edit plan: immutable regions, editable regions, room-control maps, product references, style references, and success criteria. This is where ReformAI becomes a system rather than a raw model wrapper. The planner should decide whether the request goes to Creative Mode, Locked Edit Mode, or Product/SKU Mode. This is an architectural recommendation, supported by the fact that the relevant capabilities are split across different model surfaces.

3) Generation / transformation.
Default to FLUX.2 [PRO] for high-quality editing with references. Route geometry-sensitive requests to SD 3.5 + Depth/Canny ControlNet. Keep Gemini 3.1 Flash Image, Gemini 3 Pro Image, or OpenAI GPT Image available as optional creative/reference generators for exploration, but not as the constraint backbone.

4) Local editing.
For region edits, use FLUX.1 Fill [pro] or Stability inpainting/edit endpoints. Every local edit should be driven by masks, and ReformAI should composite unchanged pixels back for protected regions rather than trusting the generator to preserve them implicitly. The model performs the local synthesis; ReformAI enforces preservation.

5) Validation and reranking.
Generate multiple candidates, then score them with a hybrid judge:
semantic fit via CLIP or a strong VLM; product/visual identity via DINOv2 embeddings; and protected-region drift via LPIPS on windows/background/untouched furniture crops. Use these scores to reject candidates that satisfy the prompt but violate constraints. This is a critical missing layer in ReformAI's current stack.

6) Commerce lane.
For SKU-grade renders, do not let the generator invent the product from scratch. Ingest product assets, place them deterministically, then use local harmonization/compositing to integrate shadows, color temperature, and scene coherence. Adobe Firefly's composite and custom-model stack is especially relevant here; research systems like AnyDoor/PowerPaint are worth testing later as specialist insert/harmonize modules.

7) Stateful session store.
Persist: original room image, accepted edit states, region masks, depth/edge maps, product references, prompt history, and validation scores. The image model APIs can support multi-turn editing, but ReformAI should own the actual state graph.

A practical pipeline is:

room photo -> segmentation/masks -> depth/edges -> edit plan -> candidate generation (4-8) -> validation/rerank -> protected-region compositing -> accepted state -> next edit

That is the architecture that supports iterative renovation workflows rather than one-shot scene generation.

F. Tradeoff Analysis

Approach 1: keep a prompt-only proprietary stack (current direction).
Visual quality can be high, latency can be low, and implementation is simple. Gemini 2.5 Flash Image is inexpensive and optimized for fast editing; OpenAI GPT Image is strong on conversational edits and production-friendly outputs. But controllability remains the weak point, especially on geometry/window preservation and selective local edits. This is the fastest path to incremental improvement, but not the path to ReformAI's stated product vision.

Approach 2: hybrid controlled editor (recommended near-term target).
This gives ReformAI the best balance of realism and controllability: FLUX.2 for strong editing quality, SD 3.5 ControlNets for room geometry, SAM 2 / Grounding DINO for masks, and a validator/reranker layer to enforce constraints. Latency will be higher than a prompt-only stack and engineering complexity will rise meaningfully, but this is the first architecture that actually matches the product requirements.

Approach 3: full commerce-grade orchestration.
This adds the product asset lane, deterministic product insertion, relighting/harmonization, and stronger validation. It gives the highest control and the only credible path to SKU-accurate shopping flows, but it is also the most complex and will cost more in both compute and implementation effort. Adobe becomes more attractive here because of composites, custom models, and enterprise/commercial posture.

Feasibility by timeframe.
In 1–2 sprints, ReformAI can absolutely add candidate generation, automatic reranking, session state persistence, and basic protected masks around windows/background/untouched furniture. It can also benchmark a better front-end editor such as Gemini 3.1 Flash Image, Gemini 3 Pro Image, or OpenAI gpt-image-2 against the current Gemini 2.5 Flash Image baseline. The full controlled pipeline with segmentation, structure control, and commerce-grade product insertion is not a 1–2 sprint change; that is a medium- to long-term program.

G. Phased Implementation Plan

Phase 1 — Immediate improvements (short-term, minimal architecture change).
Keep the current product running, but stop relying on single-shot prompting. Add 4–8 candidate generation, a vision-based reranker, and session-state persistence. Protect windows/background and preserved furniture with manual or semi-automatic masks. Benchmark Gemini 3.1 Flash Image, Gemini 3 Pro Image, and OpenAI gpt-image-2 against the current Gemini 2.5 Flash Image workflow on a fixed room-edit test suite. If staying in Google, start using the masked Imagen editing/background workflows rather than only prompt-driven Gemini edits.

Phase 2 — Controlled editing layer (medium-term).
Add SAM 2 + Grounding DINO/Florence-2 for editable/non-editable regions, depth/edge extraction, and a Locked Edit Mode that routes to SD 3.5 + ControlNet for structure-sensitive jobs and FLUX.2 for default high-quality editing. Add a real region-edit UI and start persisting mask/control artifacts per room/session. This is the phase where ReformAI stops being a prompt wrapper and becomes a controlled editing platform.

Phase 3 — Advanced orchestration and commerce.
Introduce a separate Product/SKU Mode with explicit asset ingestion, deterministic placement, local harmonization, and stronger validators for product identity and preserved regions. Add specialized insertion/inpainting modules where they materially outperform the default path. Use Adobe Firefly's product/composite stack where enterprise-safe product visualization is strategically important.

H. Final Recommendation

The clear recommendation is:

Do not architect ReformAI around any single image model.
Architect it around a multi-model constraint system.

The concrete target stack I would choose is:

Primary editor: FLUX.2 [PRO] as the default editing engine, with FLUX.2 [MAX] for premium/high-quality jobs.
Geometry-preserving lane: SD 3.5 Large + Depth/Canny ControlNet for room-locking and perspective-sensitive edits.
Segmentation/masks: SAM 2 + Grounding DINO or Florence-2.
Local edits: FLUX.1 Fill or Stability inpainting/edit endpoints.
Validation/reranking: frontier VLM judge + CLIP / DINOv2 / LPIPS.
Commerce lane: deterministic product asset insertion, with Adobe Firefly as the strongest enterprise-safe adjunct for composites/custom product models.

And just as important:

Keep Gemini/OpenAI as ideation and conversational UX layers, not as the sole constraint engine.
Do not keep Gemini 2.5 Flash Image as a prompt-only backbone. It is fine as a bridge, but not as the architecture for ReformAI's next stage.
Do not use Midjourney or Runway as the production backbone for constraint-aware renovation transforms. They are useful creative benchmarks and inspiration tools, not the core system.

The decisive answer is this: ReformAI's moat will come from the control layer, not from a monolithic model choice. If you want the platform to evolve from generative visualization into a stateful, commerce-aware renovation system, the right move is a hybrid architecture with FLUX + SD ControlNet + segmentation + validation + product compositing, not a better prompt on a single model.